FICTIONAL TEST PACKET. Not a real application.

Applicant's Name: Elena Vasquez

STATEMENT OF PURPOSE

I am applying to Northwestern MSAI to work on perception for robots that share a room with people, not on benchmark vision. In the Purdue cooperative robotics lab I was the second author on a conference paper about a gripper that uses a wrist camera. My part was the data split and the calibration code. The first author designed the policy. I can say exactly what I did not do.

I also taught a lab section of a signals course for a year. The students who struggled were usually missing linear algebra, and I started writing short notes for them. That is the teaching I want to keep doing in graduate school.

MSAI is the program I am applying to because I need the vision and learning courses next to a robotics faculty project. My plan for the year is those courses and a project on calibration under bad lighting, which is the failure mode in our paper. I have probability, linear algebra, signals, and a machine learning elective. I am applying with a specific experiment in mind.

EDUCATION

Undergraduate institution: Purdue University
Degree: B.S. Electrical Engineering, 2025
Cumulative GPA: 3.82

RESEARCH

Undergraduate researcher, Purdue cooperative robotics lab, January 2024 to May 2025. Second author: "Wrist-Camera Grasping Under Partial Occlusion," Great Lakes Robotics Meeting, 2025. I wrote the calibration code and constructed the train and test split. I did not design the policy.

PROJECTS

Calibration toolkit for the lab's wrist camera. Python and OpenCV. The current students still use the split script I wrote. It is in the lab repository, not a public project.

WORK EXPERIENCE

Laboratory teaching assistant, ECE 301 signals and systems, fall 2024 and spring 2025.

HONORS AND AWARDS

Purdue College of Engineering undergraduate research stipend, summer 2024.

LETTERS OF RECOMMENDATION

Letter 1. Professor Nina Cole, Purdue University. "Elena is the second author for a reason I can defend. She built the split that made the paper's claim measurable, and she refused to let us average away the lighting

failures. I recommend her for a research master's without reservation."

Letter 2. Dr. Paul Ortega, Purdue University. "Elena taught my signals lab. Students improved, and Elena could tell me which problem set item caused it. She is organized, precise, and ready for graduate coursework. I recommend her strongly."